

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Network Intrusion Detection]

Domain of Project – [Cybersecurity/Network Security...]

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Problem Statement -

Problem Statement : – Network Intrusion Detection

The Challenge

Create a robust network intrusion detection system (NIDS) using machine learning. The system should be capable of analyzing network traffic data to identify and classify various types of cyber-attacks (e.g., DoS, Probe, R2L, U2R) and distinguish them from normal network activity. The goal is to build a model that can effectively secure communication networks by providing an early warning of malicious activities.

The Objective

To develop an intelligent Network Intrusion Detection System (NIDS) using Machine Learning That can analyze network traffic and accurately identify normal and malicious activities

The solution should:

- To detect suspicious and malicious network activities automatically.
- To classify network traffic as Normal or Anomaly.
- To reduce manual effort in monitoring network security.
- To improve the accuracy and speed of intrusion detection using Machine Learning

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- IBM watsonx.ai Auto AI
- IBM Cloud
- Machine Learning
- Random Forest Classifier
- CSV Network Traffic Dataset
- REST API Deployment

I used IBM watsonx.ai AutoAI + IBM Watson Machine Learning (WML) + IBM Cloud + Snap Random Forest Classifier + NIDS API Deployment

Proposed Solution -

Proposed Solution: NIDS Model

The proposed solution is a Machine Learning-based Network Intrusion Detection System (NIDS) developed using IBM watsonx.ai AutoAI. The system analyzes network traffic data and classifies connections as either normal or malicious.

The dataset contains network traffic features such as protocol type, service, source bytes, destination bytes, and various connection statistics. IBM AutoAI automatically preprocesses the data, selects the best algorithm, and optimizes model performance.

After training multiple pipelines, the Snap Random Forest Classifier achieved an accuracy of approximately 99.6%. The trained model was deployed as an online API using IBM watsonx deployment services, enabling real-time intrusion detection.

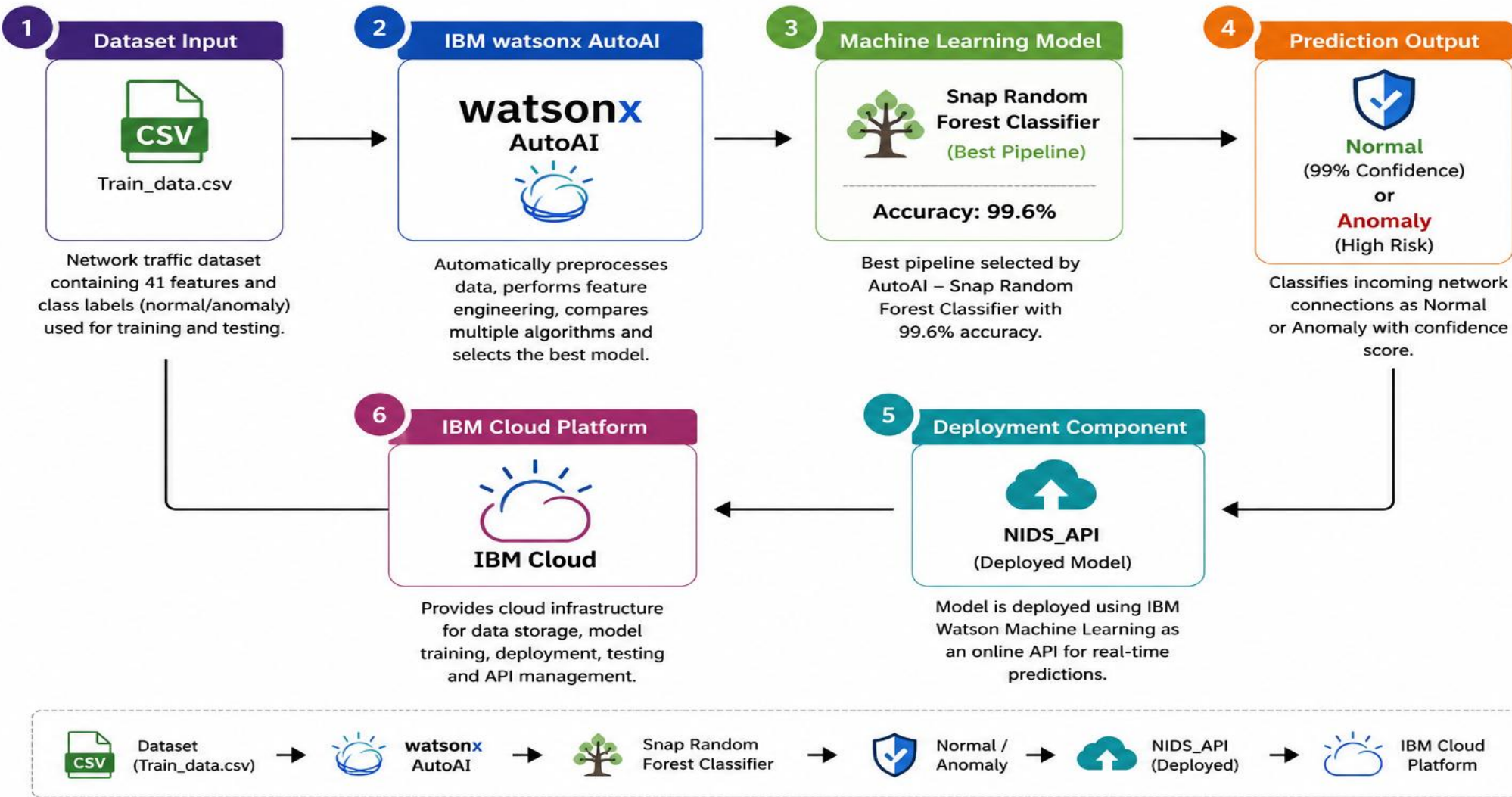
The deployed model can identify cyber attacks including Denial of Service (DoS), Probe attacks, Remote-to-Local (R2L), and User-to-Root (U2R) attacks. This solution enhances network security by providing fast and accurate threat detection.

NIDS component Used -

- 1. Dataset Input** – Network traffic dataset (Train_data.csv) containing features such as protocol type, service, source bytes, destination bytes, and connection statistics used for training and testing.
- 2. IBM watsonx AutoAI** – Automatically preprocesses data, performs feature engineering, selects the best machine learning algorithm, and optimizes model performance.
- 3. Machine Learning Model** – Snap Random Forest Classifier (Best Pipeline) selected by AutoAI, achieving approximately 99.6% accuracy for intrusion detection.
- 4. Prediction Output** – Classifies network traffic as Normal or Anomaly and provides confidence scores for each prediction.
- 5. Deployment Component** – IBM Watson Machine Learning deployment service used to deploy the trained model as an online API (NIDS_API) for real-time predictions.
- 6. IBM Cloud Platform** – Provides cloud infrastructure for model training, deployment, testing, and API management.

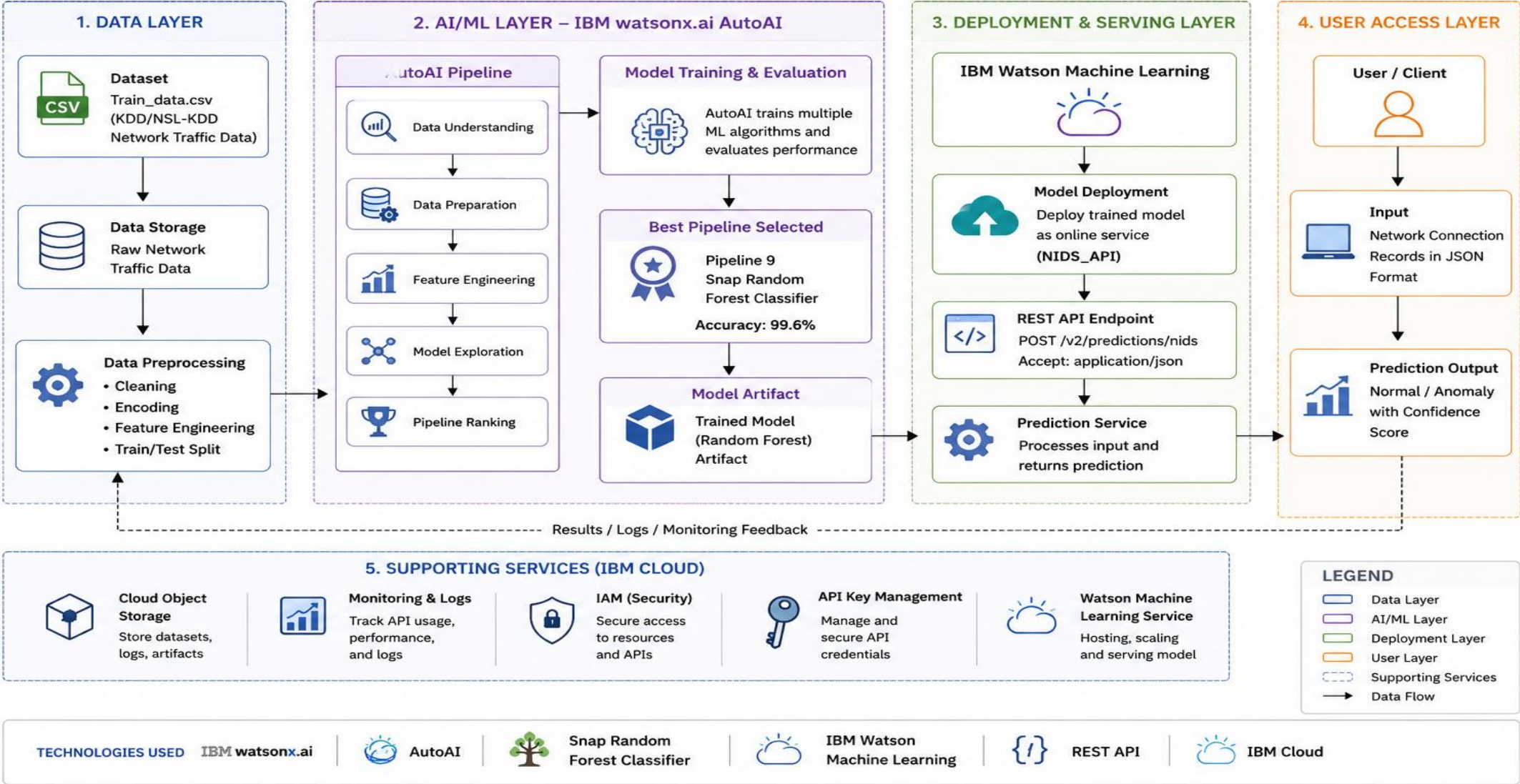
NIDS workflow -

WORKFLOW – NETWORK INTRUSION DETECTION SYSTEM



Architecture blueprint -

ARCHITECTURAL BLUEPRINT – NETWORK INTRUSION DETECTION SYSTEM



Role of Agentic AI in the solution

Role of AI in Network Intrusion Detection System:

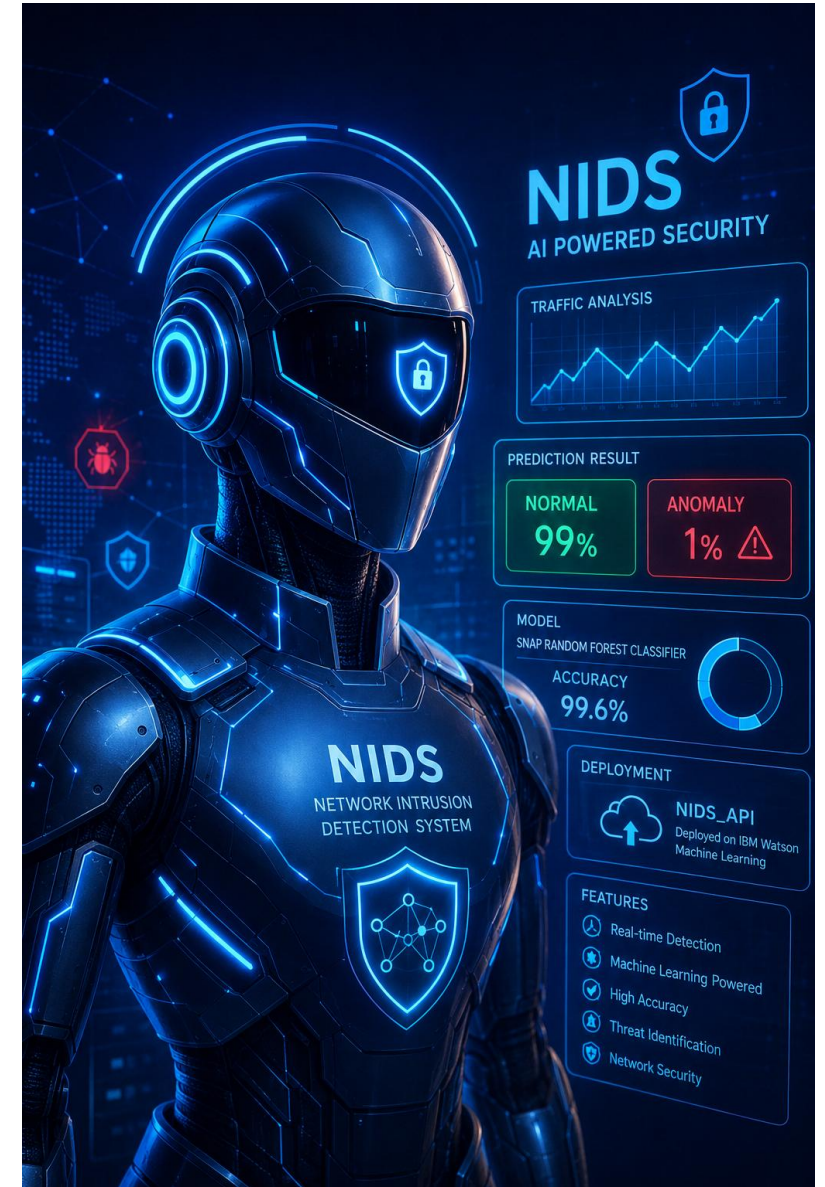
Artificial Intelligence enables the Network Intrusion Detection System (NIDS) to automatically analyze network traffic and identify malicious activities in real time. Using IBM watsonx.ai AutoAI, the system learns patterns from historical network traffic data and classifies connections as either Normal or Intrusion.

The AI model processes multiple network features such as protocol type, service, connection duration, source bytes, destination bytes, error rates, and host statistics to detect suspicious behavior. AutoAI automatically performs data preparation, feature engineering, model training, and pipeline optimization to select the best-performing algorithm.

In this project, the Snap Random Forest Classifier was selected as the optimal model, achieving approximately 99.6% accuracy. The trained model is deployed as an online API service using IBM Watson Machine Learning, allowing users to submit network traffic records and receive instant predictions.

The AI system continuously analyzes incoming network data, helping organizations detect potential cyberattacks such as unauthorized access attempts, denial-of-service attacks, and other network threats. This improves cybersecurity by reducing manual monitoring efforts and enabling faster threat detection.

Overall, AI makes the Network Intrusion Detection System intelligent, automated, accurate, and scalable, providing real-time protection against cyber threats and ensuring network security.



Technology Used

1. IBM watsonx.ai AutoAI

- Automated ML model development
- Data preprocessing, training, and model selection

2. Snap Random Forest Classifier

- Best model selected by AutoAI
- Detects Normal and Intrusion traffic
- Accuracy: 99.6%

3. IBM Watson Machine Learning

- Deploys model as an online API
- Enables real-time predictions

4. IBM Cloud

- Provides secure and scalable infrastructure
- Supports deployment and storage

5. KDD/NSL-KDD Dataset

- Network traffic dataset with 41 features
- Used for model training and testing

6. REST API

- Accepts JSON input
- Returns prediction results

7. JSON Format

- Used for sending network traffic records
- Easy integration with applications

8. Cloud Object Storage

- Stores datasets, models, and deployment artifacts

Novelty and Uniqueness

The proposed Network Intrusion Detection System (NIDS) stands out by combining Machine Learning, automated model optimization, and cloud deployment into a single intelligent cybersecurity solution.

- **AI-Powered Threat Detection** – Uses Machine Learning to automatically identify suspicious network activities and cyber-attacks from network traffic data.
- **AutoAI-Based Model Optimization** – Leverages IBM watsonx AutoAI to automatically preprocess data, select the best algorithm, and optimize model performance.
- **High Detection Accuracy** – Achieved approximately 99.6% accuracy using the Snap Random Forest Classifier, ensuring reliable intrusion detection.
- **Real-Time Prediction Capability** – The trained model is deployed as an online API on IBM Cloud, enabling real-time analysis of network traffic.
- **Detection of Multiple Attack Types** – Capable of identifying various network attacks such as DoS, Probe, R2L, and U2R along with normal network activities.
- **Scalable & Cloud-Ready Architecture** – Built on IBM watsonx.ai and IBM Cloud, making the solution suitable for enterprise-scale deployment.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)

with format – <https://github.com/bulkfracturedusk-sketch/Network-Intrusion-Detection-System>

Please test and Paste the working Github URL –

Make sure you have Uploaded following three files into your github repository

1) [app.json file](#) , 2) [yourproblemstatement.pdf file](#) 3) your projectpresntation.pptx file 4)any other files

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Future Scope-

1. Real-Time Network Monitoring Integration

- Integrate with live network monitoring tools for continuous traffic analysis.
- Enable instant detection and response to cyber threats.

2. Advanced Threat Detection Using Deep Learning

- Implement CNN, LSTM, and Transformer models.
- Improve detection of zero-day attacks and sophisticated threats.

3. Mobile & Web Dashboard

- Develop a user-friendly dashboard for real-time monitoring and alerts.
- Access intrusion reports from anywhere.

4. Automated Alert & Response System

- Send instant notifications via email, SMS, or dashboards.
- Automate actions such as blocking suspicious connections.

5. Cloud-Based Scalable Deployment

- Deploy the system across enterprise networks and cloud environments.
- Support large-scale traffic analysis with high availability.

Thank You!

Thank you for your time and interest.